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ASSIGNMENT #01

QUESTION NO-01:

$g(t) = e^{-4t} u(t)$

$e^{-4t} u(t) \Longleftrightarrow \frac{1}{4+jw}$

(a) $\Psi_g(w) = ?$

$\Psi_g(w) = |G_1(w)|^2$

$= \left(\frac{1}{|4+jw|}\right)^2$

$\Psi_g(w) = \frac{1}{16+w^2}$

(b). $E_g = ?$

$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$

$= \int_{0}^{\infty} (e^{-4t})^2 dt$

$= \int_{0}^{\infty} e^{-8t} dt$

$= \left[\frac{e^{-8t}}{-8}\right]_0^{\infty}$

$= 0 + \frac{1}{8}$

$E_g = \frac{1}{8}$ Joules

Parseval's Theorem states that the Total Energy of signal computed in the time domain equals to the total Energy computed in the frequency domain. Integrated over all frequencies and scaled by $\frac{1}{2\pi}$

$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Psi_g(w) dw$

$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{16+w^2} dw$

$\Rightarrow \int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$:

$E_g = \frac{1}{2\pi} \left[\frac{1}{4} \tan^{-1}\left(\frac{w}{4}\right)\right]_{-\infty}^{\infty}$

$= \frac{1}{8\pi} \left(\frac{\pi}{2} - \left(\frac{-\pi}{2}\right)\right)$

$= \frac{1}{8\pi} \cdot |\pi| \Rightarrow \frac{1}{8}$ Joules

This confirms that theorem holds.

1(c) _ cut-off frequency :

$$H(\omega) = \begin{cases} 2 & |\omega| \le \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$$

filter's spectral density at output is $\Psi_g(\omega) = |H(\omega)|^2 \Psi_f(\omega)$

energy

$$E_y = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} 4 \left(\frac{1}{16+\omega^2} \right) d\omega$$

$$= \frac{4}{2\pi} \left[\frac{1}{4} \tan^{-1} \left(\frac{\omega}{4} \right) \right]_{-\omega_c}^{\omega_c}$$

$$E_y = \frac{1}{2\pi} \left(2 \tan^{-1} \left(\frac{\omega_c}{4} \right) \right)$$

$$E_y = \frac{1}{\pi} \tan^{-1} \left(\frac{\omega_c}{4} \right)$$

the output signal's Energy to be exactly 90% of the input signal's total Energy ($E_y = 0.9 (E_f)$)

$$0.9 \left(\frac{1}{8} \right) = \frac{9}{80}$$

Equating E_y

$$\frac{1}{\pi} \tan^{-1} \left(\frac{\omega_c}{4} \right) = \frac{9}{80}$$

$$\tan^{-1} \left(\frac{\omega_c}{4} \right) = \frac{9\pi}{80}$$

$$\omega_c = 4 \tan \left(\frac{9\pi}{80} \right)$$

$$= 4 \tan (0.3534)$$

$$\omega_c \approx 4 (0.3693) \approx 1.477 \text{ rad/s}$$

QUESTION NO. 02 :

$$y(t) = g(t) + \alpha g(t-\tau)$$

$\alpha = 0.2$

$\tau = 5\,ys$

- Channel Transfer function

Taking Fourier Transforms on B.S using Time-shifting property

$$CF\{g(t-\tau)\} = G(w)e^{-jw\tau}):$$

$$Y(w) = G(w) + \alpha G(w)e^{-jw\tau}$$

$$= G(w)(1 + \alpha e^{-jw\tau})$$

$$H(w) = \frac{Y(w)}{G(w)}$$

$$\boxed{H(w) = 1 + \alpha e^{-jw\tau}}$$

The phase response

$$\phi_n(w) = \tan^{-1}\left(\frac{\text{imag}}{\text{real}}\right)$$

$$= \tan^{-1}\left(\frac{-\alpha \sin(w\tau)}{1 + \alpha\cos(w\tau)}\right)$$

(b). Amplitude Respons $|H(w)|$ $\phi_n(w)$

$$H(w) = 1 + \alpha((\cos(w\tau)) - j\sin(w\tau))$$

$$= (1 + \alpha\cos(w\tau)) - j(\alpha\sin(w\tau))$$

$$|H(w)| = \sqrt{(1 + \alpha\cos(w\tau))^2 + (-\alpha\sin(w\tau))^2}$$

$$= \sqrt{1 + 2\alpha\cos w\tau + \alpha^2\cos^2(w\tau) + \alpha^2\sin^2(w\tau)}$$

$\therefore \cos^2(w\tau) + \sin^2(w\tau) = 1$

$$\boxed{|H(w)| = \sqrt{1 + \alpha^2 + 2\alpha\cos(w\tau)}}$$

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(). Distortion Type & First destructive Attenuation Frequency

Because $|H(\omega)|$ is not constant and $\theta_h(\omega)$ is not perfectly linear, this multipath channel produces both amplitude distortion & phase distortion. In wireless communications, this specific periodic variation across the frequency spectrum is known as frequency-selective fading

Severe destructive attenuation (a "null or deep fade) occurs when the amplitude response $|H(\omega)|$ reaches its minimum. This happen when the cosine term is at its minimum. i.e $\cos(\omega\tau) = -1$

$$\omega\tau = \pi, 3\pi, 5\pi, \ldots$$

we get $\omega\tau = \pi$

$\omega = 2\pi f$

$$2\pi f\tau = \pi \Rightarrow f = \frac{1}{2\tau}$$

$\tau = 5\mu s = 5\times10^{-6}s$

$$f = \frac{1}{2(5\times10^{-6})} = \frac{1}{10\times10^{-6}} = \frac{1}{10^{-5}}$$

$= 100{,}000$ Hz.

$f = 100$ KHz

QUESTION NO. 03 &

(a) Power Spectral Density PSD $S_g(\omega)$

$$g(t) = 4\cos(100t) + 2\sin(300t)$$

For a periodic signal

$$g(t) = A\cos(\omega_1 t) + B\sin(\omega_1 t)$$

the two sided PSD is represented by Dirac delta functions at fundamental freq $\pm\omega_0$ and $\pm\omega_1$

$$S_g(\omega) = \frac{\pi A^2}{2}\left[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\right] + \frac{\pi B^2}{2}\left[\delta(\omega-\omega_1)+\delta(\omega+\omega_1)\right]$$

- For first term ($A=4, \omega_0=100$):
  The PSD component is $\frac{\pi 16}{2} = 8\pi$
- For second term ($B=2, \omega_1=300$):
  The PSD component is $\frac{\pi(4)}{2} = 2\pi$

$$S_g(\omega) = 8\pi\left[\delta(\omega-100)+\delta(\omega+100)\right] + 2\pi\left[\delta(\omega-300)+\delta(\omega+300)\right]$$

» Total Avg power P_t of signal.

As $P = \frac{A^2}{2}$ for sinusoids.

$$P_1 = \left(\frac{4^2}{2}\right) = \left(\frac{16}{2}\right) = 8$$

$$P_2 = \frac{4^2}{2} + \frac{2^2}{2}$$

$$= \frac{16}{2} + \frac{4}{2}$$

$$P_t = 10\ W$$

(c) Total Avg Power P_0 of the output of the system

The PSD of output is given by $S_y(\omega) = |H(\omega)|^2 S_x(\omega)$

$$|H(\omega)|^2 = \left| \frac{1}{1 + j\frac{\omega}{200}} \right|^2$$

$$= \frac{1}{1 + \left(\frac{\omega}{200}\right)^2}$$

$$P_0 = P_{100} |H(100)|^2 + P_{300} |H(300)|^2$$

At $\omega = 100$: $|H(100)|^2 = \frac{1}{1 + (100/200)^2} = 0.8$

At $\omega = 300$: $|H(300)|^2 = \frac{1}{1 + (300/200)^2} = 0.3077$

$$P_0 = 8(0.8) + 2(0.3077)$$

$$P_0 \approx 7.015 \text{ W}$$

QUESTION NO. 04 :

Criteria for Distortionless Transmission:

According to Carlson for a signal to be transmitted distortionless the o/p $y(t)$ must be an exact replica of input $x(t)$, allowing only for a constant amplitude factor (K) and constant time delay (t_d)

$$y(t) = K x(t - t_d)$$

$Y(\omega) = K X(\omega) e^{-j\omega t_d}$, meaning the ideal system transfer function is

$$H(\omega) = K e^{-j\omega t_d}$$

- Constant Amplitude Response: $|H(\omega)| = K$
- Linear Phase Response

$$\theta_h(\omega) = -\omega t_d$$

(b) Analytical Expression for Phase Delay & Group Delay

Phase Delay $t_p(\omega)$: defined as -ve phase response divided by frequency.

$$t_p(\omega) = -\frac{\theta_h(\omega)}{\omega} = -\frac{\omega t_0 - B\omega^3}{\omega}$$

$$\boxed{t_p(\omega) = t_0 + B\omega^2}$$

Group Delay ($t_g(\omega)$):

$$t_g(\omega) = -\frac{d\,\theta_h(\omega)}{d\omega} \Rightarrow \frac{d}{d\omega}(\omega t_0 + B\omega^3)$$

$$\boxed{t_g(\omega) = t_0 + 3B\omega^2}$$

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(C). Dissusion on Envelop Distortion & Physical Meaning

Envelop Distortion:

Yes the narrow-band pulse will undergo envelop distortion. Because the Group delay $t_g(\omega) = t_0 + 3\beta\omega^2$ depends quadratically on freq. (ω). the delay is not strictly constant across the signal's bandwidth. Different frequency component within the narrow pulse will travel at slightly different speeds, causing the pulse envelope to spread or broaden over it. Phenomenon of dispersion

Physical meaning:

The group delay $t_g(\omega)$ represents the time delay experienced by the envelope of a narrow-band signal centred at frequency ω. It is the effective transit time of the signal's energy through the transmission channel.

QUESTION NO. 05

a) Convert the destination SNR requirement from decibels into a numeric ratio:

$$SNR_{dB} = 10 \log_{10}(SNR_{numeric})$$

Given $SNR_{dest} = 35dB$, we solve for $SNR_{numeric}$:

$$35 = 10 \log_{10}(SNR_{numeric})$$

$$3.5 = \log_{10}(SNR_{numeric})$$

$$\Rightarrow \boxed{SNR_{numeric} = 10^{3.5} \approx 3162.28.}$$

(b) Determine the minimum required power of the received signal P_{rx} at channel output:

First calculate Noise power N within signal's bandwidth

$$B = 5kHz$$

$$\omega_m = 2\pi B$$

$$= 2\pi(5000) = 10000\pi \text{ rad/s}$$

$\Rightarrow$ Total noise power N is found by integrating the double-sided PSD $S_n(\omega)$ over $[-\omega_m, \omega_m]$:

$$N = \frac{1}{2\pi}\int_{-\omega_m}^{\omega_m} S_n(\omega)\, d\omega$$

$$N = \frac{1}{2\pi}\int_{-10000\pi}^{10000\pi} 10^{-8}$$

$$N = \frac{1}{2\pi}(10^{-8})[\omega]_{-10000\pi}^{10000\pi}$$

$$N = \frac{1}{2\pi}(10^{-8})(20000\pi)$$

$$= 10000 \times 10^{-8}$$

$$\boxed{N = 10^{-4} W}$$

Signal-to-noise Ratio (SNR) = $\frac{P_{rx}}{N}$

$$P_{rx} = SNR_{numeric} \times N$$

$$= 10^{3.5} \times 10^{-4} = 10^{-0.5} W$$

$$P_{rx} \approx 0.3162 W \ (or \ -5 dBW)$$

(c) – Minimum transmission Power P_{tx} :

The channel attenuates signal power by 20dB.

convert 20dB into Numeric loss factor (L) :

$$L_{dB} = 10 \log_{10}(L)$$

$$20 = 10 \log_{10}(L) \Rightarrow L = 10^2 = 100$$

This means transmitted power is 100 times greater than received power :

$$P_{tx} = L \times P_{rx}$$

$$P_{tx} = 100 \times 10^{-0.5}$$

$$= 100 \times 10^{-0.5} \Rightarrow 10^2 \times 10^{-0.5}$$

$$= 10^{1.5} W$$

$$P_{tx} = 31.62 W \ (or \ 15 dBW)$$

$10 \log_{10}(31.62)$